

Linke Li

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Education

Harvard T.H. Chan School of Public Health

Boston, MA

Ph.D. in Population Health Sciences, track: Genetic Epidemiology & Statistical Genetics

Aug 2026-May 2030

- Advisor: Alkes Price, PhD & Liming Liang, PhD

Duke University

Durham, NC

Master of Biostatistics | GPA: 3.94/4.0

May 2024

- Thesis: *Explainable Deep-Learning-Based Polygenic Dementia Risk Score for Alzheimer's Disease*
- Awards: Merit Scholarship Recipient, Duke University (2022–2024)

University of California, Davis

Davis, CA

Bachelor of Animal Science

March 2022

- Awards: Dean's List (2021); Xu Family Endowed Scholarship (2021–2022)

Publications & Preprints

*Co-first authors, #Co-senior authors

1. **Linke Li***, Seyedeh Zekavat*, Xinyu Wang, Buu Truong, Alexander Bick, et al., Pradeep Natarajan*, Zhi Yu*. **Multi-ethnic Transcriptomic Analysis of Clonal Hematopoiesis Reveals Driver-Specific Pathways and Causal Associations.** *Manuscript in preparation.*
2. **Linke Li***, Ahmed Alaa*, Youxin Tan, Ilker Demirel, et al., Pradeep Natarajan#, Zhi Yu#. **Machine learning cross-platform proteomic imputation enables protein quality scoring and replication of epidemiological associations.** *Manuscript under peer review at Nature Communications.*
3. **Linke Li**, Yu Zhang, Buu Truong, Zhi Yu, et al., Michael Honigberg. **Endothelial Susceptibility-Related Genetic Variants and Hypertensive Disorders of Pregnancy.** *Manuscript accepted at Arteriosclerosis, Thrombosis, and Vascular Biology.*
4. Ramesh Kumar*, **Linke Li***, Sarah Urbut, et al., Reuben Kapur#, Zhi Yu#. **Inflammatory Bowel Disease Augments Clonal Hematopoiesis of Indeterminate Potential.** *Blood*, doi: [10.1182/blood.2025032339](https://doi.org/10.1182/blood.2025032339).
5. Santhosh Pasupuleti*, **Linke Li***, Rahul Kanumuri, et al., Zhi Yu#, Reuben Kapur#. **Inhibition of IL-17A/GLP-1 axis suppresses obesity-induced early-onset leukemia driven by oncogenic PTPN11 mutation.** *Manuscript accepted at Journal of Clinical Investigation.*
6. **Linke Li**, Jing Tian, Gary Beecham, et al., Sheng Luo, Da Ma. **Explainable Deep-Learning-Based Polygenic Dementia Risk Score for Alzheimer's Disease.** *Manuscript in preparation.*
7. Zhi Yu, Amélie Vromman, et al., **Linke Li**, et al., Pradeep Natarajan#. **Human Plasma Proteomic Profile of Clonal Hematopoiesis.** *Nature Communications*, doi: [10.1038/s41467-025-66755-z](https://doi.org/10.1038/s41467-025-66755-z).
8. Niekbachsh Mohammadnia, **Linke Li**, et al., Honigberg#, El Messaoudi#. **Lipoprotein(a), Inflammatory Biomarkers, and Risk of Coronary Artery Disease and Aortic Valve Stenosis.** *Manuscript accepted at JAMA Cardiology.*
9. Daniel Ezzat, Diana M Lopez, Brian L Claggett, **Linke Li**, Niekbachsh Mohammadnia, Art Schuermans, Jan Hemeryck, Annie Chang, Samantha Murillo, Michelle L O'Donoghue, Behnood Bikdeli, Zhi Yu, Pradeep Natarajan, Aniruddh P Patel, Maria A Pabon#, Michael C Honigberg#. **Lipoprotein(a) and incident venous thromboembolism in pre- and postmenopausal women, and in men.** *European Heart Journal*, doi: [10.1093/eurheartj/ehag252](https://doi.org/10.1093/eurheartj/ehag252).
10. Yixuan Liu, **Linke Li**, et al., Honigberg#, Zhi Yu#. **Association of Life's Essential 8 with Clonal Hematopoiesis and Coronary Artery Disease in the UK Biobank.** *Manuscript in revision at JACC: Cardiooncology.*
11. Shengxin Liang*, Min Seo Kim*, Yang Sui, Youxin Tan, **Linke Li**, et al., Akl C. Fahed#, Zhi Yu#. **Evaluating Individual Level Performance of Polygenic Risk Scores Using Early Onset Unexpected Coronary Artery Disease as a Benchmark.** *Manuscript accepted at Nature Communications.*

12. Jacqueline Dron, Michael Pan, et al., **Linke Li**, et al., Pradeep Natarajan. **Integrative multi-omic analysis of regional fat depots reveals molecular determinants of human fat distribution.** *Preparing for resubmission at Circulation.*
13. Cesar De Jeronimo Diaz*, Huajun Liao*, Brian Hardaway, **Linke Li**, et al., Trevor P. Fidler. **Ppm1d Clonal Hematopoiesis Does Not Promote Atherosclerosis in Murine Models.** *Manuscript accepted at Arteriosclerosis, Thrombosis, and Vascular Biology.*
14. Micayla Flores, Yunong Zhao, **Linke Li**, Wesley A. Spacht, Charles Kooperberg, JoAnn Manson, Charles Eaton, Alexander Reiner, Michael Honigberg, Emily Lau **Protein biomarkers of infertility implicate inflammatory, cell migration and adhesion, and vascular regulation pathways.** *Manuscript accepted at JACC: Advances.*
15. Yang Sui, Shinwan Kany, Shaan Khurshid, Kelvin Supriami, Joel Rämö, Sean Jurgens, Nobuyuki Enzan, Seung Hoan Choi, Zhi Yu, **Linke Li**, Buu Truong, Xingyu Chen, Min Seo Kim, Injeong Shim, Raphael Twerenbold, James Pirruccello, Pradeep Natarajan, Patrick Ellinor, Akl Fahed **Genetically Mediated Differences in LDL Cholesterol and Risk of Venous Thromboembolism.** *Manuscript in revision at JACC.*

Presentations & Talks

Platform (Oral) Presentation:

- ASHG 2025 Annual Meeting: Multi-ethnic Transcriptomic Analysis of Clonal Hematopoiesis Reveals Driver-Specific Pathways and Causal Associations.

Moderated Poster Session / Poster Blitz:

- AHA 2025 Scientific Session: Endothelial Susceptibility-Related Genetic Variants and Hypertensive Disorders of Pregnancy.
- CHARGE Meeting 2025: Multi-ethnic Transcriptomic Analysis of Clonal Hematopoiesis Reveals Driver-Specific Pathways and Causal Associations.

Poster Presentation:

- Alzheimer's Association International Conference 2024: Deep-learning-based Polygenic Dementia Risk for Alzheimer's Disease.

Contributed / Co-authored Presentations:

- Xinyu Wang, **Linke Li**, et al. CHIP: A Modifiable Risk Factor for Prostate and Bladder Cancer. *Lightning Talk, ASHG 2025.*
- Zhi Yu, Amélie Vromman, **Linke Li**, et al. Causal Proteomic Consequences of Clonal Hematopoiesis. *Moderated Poster, AHA Scientific Session 2025.*
- Niekbachsh Mohammadnia, **Linke Li**, et al. Lipoprotein(a), Inflammatory Biomarkers, and Risk of CAD and Aortic Valve Stenosis. *Poster, ESC Heart & Stroke 2025 / EuroThrombosis 2025.*
- TJ Yao, **Linke Li**, et al. Developmental Enamel Defects in Young Adults with Perinatal HIV Infection. *Poster, IADR 2025.*

Research Experience

Massachusetts General Hospital & Broad Institute of MIT and Harvard

Computational Associate

Advisors: Zhi Yu, M.B., Ph.D. & Michael Honigberg, M.D., M.P.P.

Dec 2024 - Present

Lead Projects:

Transcriptomic Profile of Clonal Hematopoiesis of Indeterminate Potential (CHIP)

- Developed and optimized custom transcriptomic data processing (QC, normalization, PEER factors) pipelines for **Bulk RNA-Seq data** from 5 **NHLBI multi-ethnic TOPMed** cohorts (MESA, FHS, CARDIA, COPDGene, WHI), referencing the GTEx RNA-seq pipeline.
- Assessed correlations between transcripts and CHIP carriers with **linear mixed effects models** incorporating a kinship matrix in each cohort.
- Batched 80,000+ transcript-CHIP association analyses per cohort using **GNU Parallel** and **array jobs** on HPC clusters, significantly improving computational efficiency.
- **Meta-analyzed** cohort results stratified by tissue type (whole blood & PBMC); validated causal transcript-CHIP

signals with **bidirectional Mendelian Randomization (MR)**.

- Integrated findings with proteomic and DNA methylation data via **pathway enrichment analysis** (IPA, Gene Ontology, KEGG, STRING).

Related Work: Human Plasma Proteomic Profile of Clonal Hematopoiesis

- Optimized computational efficiency of the bidirectional MR pipeline (UK Biobank Pharma Proteomics Project, UKB-PPP); established it as the reference two-sample MR pipeline for the group.

Endothelial Susceptibility-Related Genetic Variants and Hypertensive Disorders of Pregnancy

- Investigated shared genetic architecture between hypertensive disorders of pregnancy (HDP) and coronary artery disease (CAD) via **pathway-specific polygenic risk scores (PRS)**.
- Extracted variant data from UK Biobank genotypes using **Bgenix**, calculated PRS with **PLINK2**, and validated associations with two-sample MR.

Inflammatory Bowel Disease Augments Clonal Hematopoiesis of Indeterminate Potential

- Assessed IBD-CHIP association using Cox proportional hazard models and two-sample MR in the *UK Biobank*; replicated findings in *All of Us*.

Targeting GLP-1R/IL-17A Axis in Obesity-Induced Leukemia (*PTPN11* Mutations)

- Extracted WES variant data in **UK Biobank RAP**; conducted Cox PH, *t*-tests, and Fisher exact tests on *PTPN11* germline mutations, obesity/diabetes, and myeloid malignancies.
- Used **GeneBass** for rare-variant analyses (**SKAT-O** and **Burden tests**) to validate murine findings.

Calibration of SomaScan and Olink Protein Measurements

- Benchmarked multiple ML methods (Isotonic Regression, XGBoost, LightGBM, Hist-Gradient Boosting) in Python for large-scale proteomic harmonization (UK Biobank, TOPMed).
- Utilized Hist-Gradient Boost to the MESA cohort to harmonize SomaScan and Olink measurements.

Other Responsibilities:

- Mentoring ~7 students (visiting medical students, residents, fellows, and master's students).
- Documented and established onboarding / standard operating procedures for HPC cluster usage.
- Built pipelines for batch curation and formatting of phenotypes & baseline drug usage in UK Biobank.

Harvard T.H. Chan School of Public Health

July 2024 - Dec 2024

Center for Biostatistics in AIDS Research (CBAR) | PHACS Network

Developmental Enamel Defects in Young Adults with Perinatal HIV Infection

- Authored Analysis Reports using SAS to model complex, correlated clinical outcomes with **generalized estimating equations** and **generalized linear mixed models**.

Duke University School of Medicine, Department of Biostatistics & Bioinformatics

M.S. Thesis Advisors: Da Ma, Ph.D. (Wake Forest); Sheng Luo, Ph.D. (Duke)

Explainable Deep-Learning-Based Polygenic Dementia Risk Score for Alzheimer's Disease

- Performed GWAS and PRS analysis on ADNI genetic/clinical data; applied QC, imputation, and cross-validation to identify high-risk SNPs with model explainability via **Captum**.
- Developed a **1D-CNN (PyTorch)** for AD classification on SNP dosage data; achieved interpretability with occlusion and guided backpropagation.

Skills & Certifications

Programming & Analysis: R, SAS, Python, High-Performance Computing (HPC) clusters

Genomics & Bioinformatics: Anaconda3, PLINK2, bcftools, htlib, samtools, BGENIX, MAGMA, Picard/LiftOver, GWAMA, LDSC, PRS-CS, PRSice

Certifications: SAS Certified Professional — (1) Clinical Trials, (2) Advanced, (3) Base Programming using SAS 9.4

Teaching & Work Experience

Dizal Pharma | Statistical Programmer

Shanghai, China | May – Aug 2023

- Identified a critical programming logic flaw in an existing SDTM macro; communicated findings to senior

management, resulting in full SAS macro reconstruction for future FDA submissions.

- Assimilated CDISC Standards and FDA requirements (SEND, SDTM, ADaM); developed a QC rulebook and implemented SAS macros to detect multiple data tabulation errors, delivered as QC reports.

University of California, Davis | Student Academic Tutor

Davis, CA | Nov 2021 – Mar 2022

Academic Assistance and Tutoring Centers

- Tutored BIS101 (Genes & Gene Expression), PLS120 (Applied Statistics for Agriculture), MAT16C (Short Calculus).
- Hosted weekly office hours and on-demand 1-on-1 appointments for students and student-athletes.